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Xmodel-LM Technical Report

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Abstract

We introduce Xmodel-LM, a compact and efficient 1.1B language model pretrained on around 2 trillion tokens. Trained on our self-built dataset (Xdata), which balances Chinese and English corpora based on downstream task optimization, Xmodel-LM exhibits remarkable performance despite its smaller size. It notably surpasses existing open-source language models of similar scale. Our model checkpoints and code are publicly accessible on GitHub at `https://github.com/XiaoduoAILab/XmodelLM`.

1 Introduction

Large language models (LLMs) have demonstrated remarkable performance across various natural language tasks, often requiring minimal examples of natural language instructions [Brown et al., 2020], thereby alleviating the necessity for extensive feature engineering.

At the heart of this progress lies language modeling, typically formulated as an unsupervised distribution estimate from sequences of symbols $(s_1, ..., s_{n-1})$ within a variable-length sequence x. Leveraging the natural sequential order of languages, the joint probability over symbols is commonly factorized into a product of conditional probabilities:

$$p(x) = \prod_{i=1}^{n} p(s_n | s_1, ..., s_{n-1})$$

Recent advancements in natural language processing (NLP) have largely stemmed from the scaling up of language model sizes, driven by the observed functional relationship between model performance and size [Henighan et al., 2020]. However, the accompanying rise in operational costs poses a significant hurdle to their widespread adoption.

In this paper, we introduce Xmodel-LM, a compact vision-language assistant powered by a relatively small language model. Remarkably, Xmodel-LM achieves performance comparable to state-of-the-art models of similar scale across numerous LLM benchmark tests, showcasing its potential for a wide array of practical tasks.

2 Pretraining

This chapter details the pretraining process of Xmodel-LM. First, we introduce the sources and composition of our corpus, as well as our preprocessing methods. Second, we describe the construction of our customized tokenizer. Finally, we detail the model architecture and training parameter configurations.

2.1 Training Data

Data sourcing: In the process of constructing the training corpus and allocating weights, our primary objective is to ensure the quality and diversity of the training data. Our original dataset primarily consists of aggregated training data from other LLMs, such as Redpajama [Computer, 2023], subsets of the Pile [Gao et al., 2020] and StarCoder [Li et al., 2023]. To address deficiencies in the distribution of book and mathematical data within the training data distribution, we have also incorporated FanFics[1] and OpenWebMath [Paster et al., 2023]. Additionally, we have added the Chinese data source PTD [Wang et al., 2024] and WanJuan [He et al., 2023] to imbue our model with a certain level of proficiency in Chinese.

Data processing: We are committed to ensure the quality of the data and reducing its redundancy. We first employ heuristic methods such as paragraph length and punctuation ratio for initial filtering. Subsequently, we utilize a 5-gram Kneser-Ney model based on KenLM Library [Heafield, 2011] to compute text perplexity for further quality filtering. In the next stage, we employe a locality-sensitive hashing method based on SimHash to deduplicate the training data. To balance deduplication quality and efficiency, we implement a bucketing strategy on the entire dataset, enabling the deduplication process to scale efficiently across large datasets. Finally, we tokenize all datasets using our custom-trained tokenizer, and designed different sampling weights for the datasets based on their characteristics, as shown in Table 1.

Data Source	Dataset	Num Tokens	Sampling Weight	Epochs	Category	Language
Redpajama	Arxiv	31,336,679,261	0.0160	1	Academic	English
	Book	29,633,248,538	0.0300	1	Book	English
	C4	192,696,661,887	0.1000	1	Web	English
	Common Crawl	1,251,868,330,446	0.5600	0.88	Web	English
	Github	59,063,773,003	0.0150	0.5	Code	English
	Stackexchange	22,728,030,774	0.0174	1.5	Social	English
	Wikipedia	34,312,919,854	0.0520	3	Academic	English
Pile	BookCorpus	562,392,085	0.0006	2	Book	English
	PubMed	17,698,877,602	0.0100	1	Academic	English
AMPS	AMPS	269,936,326	0.0003	2	Math	English
FanFics	FanFics	1,803,437,344	0.0020	2	Book	English
OpenWebMath	OpenWebMath	7,150,335,312	0.0080	2	Math	English
StarCoder	StarCoder	306,812,862,958	0.0536	0.3	Code	English
WanJuan	Law	9,080,387,832	0.0100	2	Academic	Chinese
	News	5,175,531,875	0.0050	2	Academic	Chinese
	Patent	4,559,904,057	0.0050	2	Academic	Chinese
	Webtext	126,429,462,230	0.0644	1	Web	Chinese
PTD	PTD	165,879,069,486	0.0507	0.6	Web	Chinese

Table 1: Detailed composition of the training set.

2.2 Tokenizer

We employ the unigram algorithm [Kudo, 2018] for data tokenization, utilizing the implementation provided by Sentence-Piece [Kudo and Richardson, 2018]. In contrast to the extensive vocabularies used in prevailing open-source models, our tokenizer is trained on a mixed corpus of Chinese and English, with a vocabulary size of only 32,000. The comparison of the Xmodel-LM tokenizer with other tokenizers is shown in the Table 2. Despite its small size, our tokenizer demonstrates impressive compression rates on test data.

The Xmodel-LM tokenizer is trained using a subset of the Xmodel-LM pre-training corpus, without any applied text normalization. To improve the encoding of numeric data, numbers are split into

[1]https://huggingface.co/datasets/marianna13/fanfics

Tokenizer	Vocab Size	Compression Rate ↓
LLaMA 2	32,000	0.7524
InternLM 2	103,168	0.4124
Baichuan 2	125,696	0.4103
Xmodel-LM	32,000	**0.3841**

Table 2: Comparison of vocabulary size and text compression rate of Xmodel-LM's tokenizer with other models. Lower values indicate better compression.

individual digits. Character coverage is set to 0.9999, with rare characters being represented by UTF-8 bytes. Additionally, we set the maximum token length to 16 to accommodate Chinese phrases.

2.3 Model architecture

We adopt a similar model architecture to LLama 2 [Touvron et al., 2023] with the following details:

Hidden size	Intermediate size	Attention heads	KV heads	Layers	Context Len
2048	5632	32	4	24	4096

Table 3: Detailed settings of Xmodel-LM.

Rotary Positional Embedding. We integrate rotary positional embeddings (RoPE) [Su et al., 2023] at each layer of the network.

RMSNorm. To enhance training stability, we utilize the RMSNorm [Zhang and Sennrich, 2019] function to normalize the input of each transformer sub-layer, without normalizing the output. Linear layers do not incorporate bias, and word embeddings are not tied.

SwiGLU. We replace the conventional ReLU non-linearity with the SwiGLU [Shazeer, 2020] activation function to optimize performance.

Grouped-query attention. For efficient training and inference, we employ grouped-query attention (GQA) [Ainslie et al., 2023], featuring 32 attention heads and 4 KV heads.

2.4 Training

Training is conducted on a single node equipped with 7×H800 GPUs. To enhance training efficiency and boost Model FLOPS Utilization (MFU), we employ Distributed Data Parallel (DDP) and FlashAttention-V2.

We utilize the cumulative gradient updating method with a mini-batch size of 4 and a gradient accumulation step of 30 per GPU, resulting in a global batch size of 840 with a sequence length of 4096. This setup yields a total token count of 3,440,640 per iteration. Training spans 600,000 iterations, accumulating to a total token count of 2,064,384,000,000.

For optimization, we employ the AdamW optimizer with a maximum learning rate of 6e-4. The learning rate undergoes linear increase from zero to the maximum over the first 2000 updates, followed by annealing to 6e-5 using a cosine schedule. The batch size is set to around 3.5M tokens, with weight decay assigned as 0.1. Additionally, we apply a gradient clipping threshold of 1.0 to regulate the gradient value.

Refer to the training log in the Figure 1, which includes trend graphs showing the training and validation losses as the training token count increases. We use OpenWebText [Gokaslan et al., 2019], which is not included in the training set, as the validation set to calculate the validation loss.

3 Results

Baseline models For the sake of comparability, we chose several recently popular decoder-only architecture models with a scale of around 1 billion parameters for comparison. Specifically, we

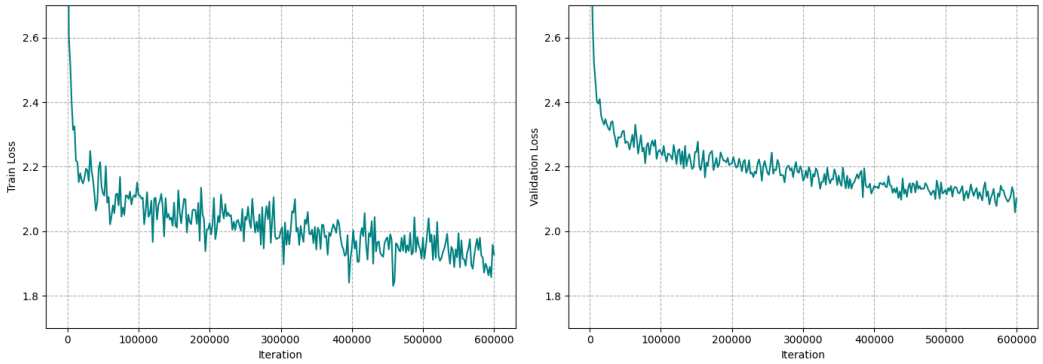

Figure 1: The trend of training and validation loss during pretraining.

compare Xmodel-LM with OPT [Zhang et al., 2022], Pythia [Biderman et al., 2023], TinyLLaMA [Zhang et al., 2024], MobileLLaMA [Thawakar et al., 2024], H2O-danube [Singer et al., 2024], InternLM2 [Cai et al., 2024] and Qwen1.5 [Bai et al., 2023].

Commonsense reasoning tasks For evaluating our models we use the Language Model Evaluation Harness [Gao et al., 2023]. Specifically, our evaluation comprised: ARC-Challenge [Clark et al., 2018b], ARC-Easy [Clark et al., 2018b], Boolq [Clark et al., 2019], HellaSwag [Zellers et al., 2019], OpenBookQA [Mihaylov et al., 2018], PiQA [Bisk et al., 2019], SciQ [Welbl et al., 2017], TriviaQA [Joshi et al., 2017], Winogrande [Sakaguchi et al., 2021]. To ensure evaluating fairness and reproducibility, all evaluation metrics reported in our study are measured in the same environment. It is worth mentioning that we used the raw accuracy metrics, rather than the normalized ones. The evaluation results are presented in Table 4, all models are evaluated in a zero-shot setting on these tasks. We notice Xmodel-LM outperforms several baseline models, particularly surpassing TinyLlama in various evaluation metrics, and it's comparable to Qwen1.5 in terms of overall performance.

Model	**ARC-c**	**ARC-e**	**Boolq**	**HS.**	**OB.**	**PiQA**	**SciQ**	**TQ.**	**Wino.**	**Avg**
OPT-1.3B	23.29	57.03	57.80	41.52	23.20	71.71	84.30	7.48	59.59	47.32
Pythia-1.4B	25.60	57.58	60.34	39.81	20.20	71.06	85.20	5.01	56.20	47.00
TinyLLaMA-3T-1.1B	27.82	60.31	57.83	44.98	21.80	73.34	88.90	11.30	59.12	48.59
MobileLLaMA-1.4B	26.28	61.32	57.92	42.87	23.60	71.33	87.40	12.02	58.25	49.00
Qwen1.5-1.8B	32.25	64.69	66.48	45.49	23.80	73.45	92.90	1.01	61.17	51.25
H2O-danube-1.8B	32.94	67.42	65.75	50.85	27.40	75.73	91.50	25.05	62.35	55.44
InternLM2-1.8B	37.54	70.20	69.48	46.52	24.40	75.57	93.90	36.67	65.67	57.77
Xmodel-LM-1.1B	28.16	62.29	61.44	45.96	24.00	72.03	89.70	18.46	60.62	51.41

Table 4: Performance on commonsense reasoning tasks. Models marked in green perform worse than Xmodel-LM, while models marked in red perform better than Xmodel-LM.

Problem-solving evaluation For exploring the performance of the model beyond common-sense reasoning, we also evaluate the model's problem-solving capability. Specifically, our evaluation comprised:

- BIG-Bench Hard (BBH) [Suzgun et al., 2022]: this is a subset of 23 challenging tasks from the BIG-Bench benchmark [Srivastava et al., 2023] designed to gauge the proficiency of a language model in comprehending and executing complex instructions.
- The General Language Understanding Evaluation (GLUE) [Wang et al., 2018]: this is a collection of resources for training, evaluating, and analyzing natural language understanding systems.
- Grade School Math 8K (GSM8k) [Cobbe et al., 2021]: this comprises 8.5k high-quality grade school math word problems, diverse in linguistic aspects. It was curated specifically for question answering tasks involving multi-step reasoning on basic mathematical problems.

- Massive Multitask Language Understanding (MMLU) [Hendrycks et al., 2021]: this is used to measure a model's breadth of world knowledge and its problem-solving proficiencies across diverse subjects.

The evaluation results are presented in Table 5. It can be observed that the Xmodel-LM has achieved the highest performance on BBH compared to the baseline models, and it demonstrates strong competitiveness overall.

Model	**BBH** 3-shot	**GLUE** 5-shot	**GSM8K** 5-shot	**MMLU** 5-shot	**Avg**	**Avg** w.o. GSM8k
OPT-1.3B	22.67	51.06	0.83	26.70	25.32	33.48
Pythia-1.4B	25.37	52.23	1.63	25.40	26.16	34.33
MobileLLaMA-1.4B	23.48	43.34	1.44	24.60	23.22	30.47
TinyLLaMA-3T-1.1B	26.75	48.25	1.97	25.70	25.67	33.57
H2O-danube-1.8B	27.31	49.83	1.90	25.70	26.19	34.28
InternLM2-1.8B	16.86	58.96	23.50	42.00	35.34	39.27
Qwen1.5-1.8B	13.84	64.57	33.59	45.10	39.28	41.17
Xmodel-LM-1.1B	27.34	52.61	2.58	25.90	27.11	35.28

Table 5: Performance on problem-solving tasks. Models marked in green perform worse than Xmodel-LM, while models marked in red perform better than Xmodel-LM.

Chinese ability Besides evaluating the model's English proficiency, we also conducted assessments on its Chinese language capabilities due to the presence of a certain proportion of Chinese content in our corpus. Specifically, our evaluation comprised:

- ARC [Clark et al., 2018a], this consists of 7,787 science exam questions drawn from a variety of sources, including science questions provided under license by a research partner affiliated with AI2. We utilized the Chinese translation version of the original work in our study.
- XCOPA [Edoardo M. Ponti and Korhonen, 2020], this is designed to assess how well machine learning models can transfer commonsense reasoning across different languages. It is a translated and reannotated version of the English COPA [Gordon et al., 2011] and includes 11 languages from 11 different language families and various regions worldwide.
- XNLI [Conneau et al., 2018], this is designed to evaluate the ability of natural language processing systems to understand and transfer knowledge across multiple languages.

The evaluation results are presented in Table 6, all models are evaluated in a zero-shot setting on these tasks. We observed that by adding 15% Chinese tokens, our model gained a certain degree of understanding and generation capabilities in Chinese, surpassing some existing models, but still weaker compared to InternLM2 and Qwen1.5.

Model	**ARC-zh**	**XCOPA-zh**	**XNLI-zh**	**Avg**
OPT-1.3B	18.80	53.00	33.45	35.08
Pythia-1.4B	21.03	52.60	34.06	35.90
MobileLLaMA-1.4B	20.26	52.80	33.82	35.63
TinyLLaMA-3T-1.1B	21.37	56.80	33.25	37.14
H2O-danube-1.8B	21.79	55.60	34.74	37.38
InternLM2-1.8B	27.69	66.80	34.58	43.00
Qwen1.5-1.8B	32.14	66.00	39.28	45.81
Xmodel-LM-1.1B	26.24	60.60	36.02	40.95

Table 6: Performance on Chinese tasks. Models marked in green perform worse than Xmodel-LM, while models marked in red perform better than Xmodel-LM.

4 Case study

4.1 Evolution of model's performance

We tracked and recorded the model's performance on the common-sense reasoning tasks during the pretraining process, as shown in Figure 2. It can be observed that the performance of Xmodel-LM improves as training progresses, surpassing TinyLLaMA in multiple tasks. Moreover, we observe an approximate linear link between log iteration steps and model metrics gains across most tasks.

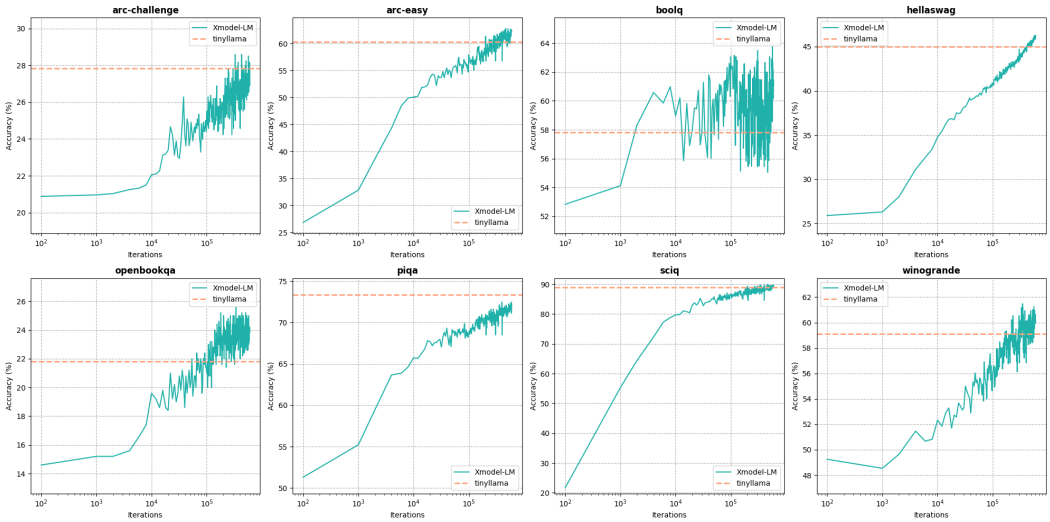

Figure 2: Evolution of performance in commonsense reasoning tasks during pre-training

4.2 Generalization vs memorization

Recent advancements in mechanical interpretability have uncovered a phenomenon known as superposition, which can represent more features than expected in neural networks. During the pre-training of Xmodel-LM, we observed shifts in the L_2-norm ($\|\theta\|_2$) of the model parameters, mirroring trends identified in prior research [Tom Henighan, 2023]. Specifically, the training process can be roughly delineated into three stages, as depicted in Figure 3.

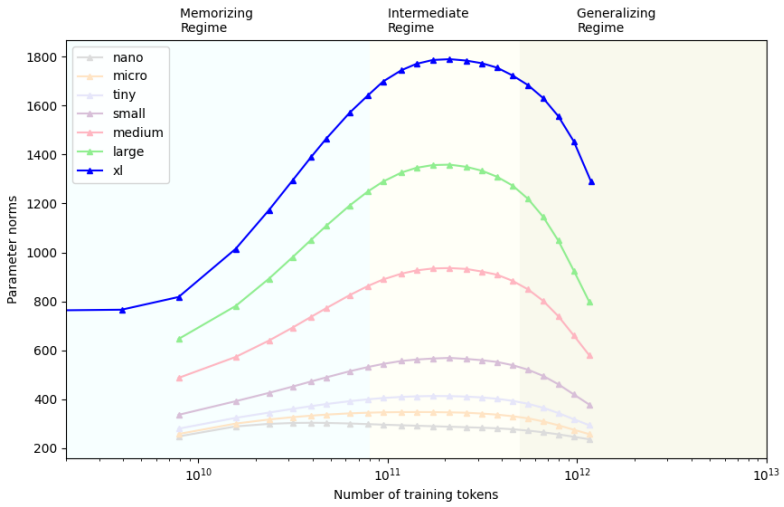

Figure 3: Shifts in the L_2-norm of parameters during pre-training

This replication underscores a similar tripartite division, the model is in a memorization state when the amount of training data is small, during which the L_2-norm of the model gradually increases. After passing through an intermediate state, the model transitions to a generalization state, during which the L_2-norm of the model gradually decreases.

5 Conclusions

In summary, our work involves training a 1.1B-sized language model (Xmodel-LM) on a self-built tokenizer and corpus containing both Chinese and English data. Our model demonstrates competitive performance on evaluation datasets compared to models of similar scales. Our efforts contribute to the advancement of knowledge by showcasing the potential of training smaller models with extensive datasets for future applications.

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